ANINDYA MIDHEY

+91 7044908773 | anindyamidhey2019@gmail.com | LinkedIn | GitHub

SUMMARY

My experience combines a strong foundation in Mathematics, Data Science, and Artificial Intelligence with hands-on work in Machine Learning, Deep Learning and NLP, Computer Vision and Reinforcement Learning. I am comfortable understanding algorithms and applying them to build practical AI and data-driven systems using Python, SQL, PyTorch, OpenCV, Scikit-Learn, OCR, and RAG.

SKILLS

- **Languages:** Python, C Programming, C++, SQL, LaTeX
- **ML/AI:** Machine Learning, Artificial Intelligence, Deep Learning, NLP, Computer Vision, Reinforcement Learning
- **Frameworks:** OpenCV, PyTorch, Scikit-Learn, NumPy, Pandas, Seaborn, Matplotlib
- **Operating Systems:** Windows, Linux (Ubuntu)
- **Tools:** Git/GitHub, MS Office, Jupyter Notebook, Google Colab, VS Code, MySQL

PROJECTS

Semantic Kinect Foam – 3D Scene Reconstruction and Object Extraction
RKMVERI | Ongoing

- Working on 3D scene reconstruction using real-world RGB-D data captured with the Azure Kinect DK.
- Using COLMAP to estimate camera poses and generate 3D points from multiple images.
- Exploring SAM and DEVA for object masking and consistent object segmentation across multiple views.
- Integrating Vision-Language Models (VLMs) to assign meaningful labels to objects and scene elements.
- Using Kinect depth data to study real-world scale, spatial relationships, and challenging objects such as glass.

Real-Time Facial Emotion Recognition |
RKMVERI | Feb 2026 – Apr 2026

- Developed a real-time facial emotion recognition system using CNN-based deep learning models for webcam-based multi-class emotion classification.
- Integrated Haar Cascade and MTCNN for face detection and preprocessing, and optimized the inference pipeline for low-latency prediction under varying lighting and facial orientations.

Project Akshar – Document Intelligence and RAG System |
RKMVERI | Feb 2026 – Apr 2026

- Built an end-to-end document question-answering system integrating image preprocessing, OCR, semantic embeddings, vector search, and Retrieval-Augmented Generation (RAG).
- Implemented semantic retrieval for multi-page documents and bounding-box based evidence highlighting to connect generated answers with relevant document regions.

Distributed Gaming Analytics System |
RKMVERI | Feb 2026 – Apr 2026

- Developed a distributed CS2 player analytics and performance prediction system using Machine Learning and Ray-based parallel processing.
- Performed feature engineering using kills, assists, ADR, KAST, KD difference, and other player-performance statistics and compared predictive models.

Laptop Price Prediction – Regression Analysis |
RKMVERI | Sep 2025 – Nov 2025

- Built a regression pipeline to predict laptop prices from RAM, processor, storage, GPU, display, and other hardware specifications.
- Performed preprocessing, feature engineering, categorical encoding, exploratory data analysis, and model evaluation using MAE, MSE, and R^2.

EDUCATION

M.Sc. in Data Science and Artificial Intelligence **[Ongoing]**
Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI), Howrah
CGPA: 5.61 (Sem 1)

B.Sc. (Honours) in Mathematics **2021 – 2024**
City College, Kolkata
CGPA: 6.466

COURSEWORK

Machine Learning, Artificial Intelligence, Deep Learning, Computer Vision, Probability and Statistics, Linear Algebra, Time Series Analysis, Data Structures and Algorithms, Distributed Computing, Programming in Modern C++

ACHIEVEMENTS & CERTIFICATIONS

GATE 2026 (DA): Qualified | **IIT JAM 2025 (MA)**: AIR 1424 | **CUET-PG 2025 (Mathematics)**: Score 124 | **NPTEL:** Programming in Modern C++ (2025) – `Elite` | **CITA:** Certificate in Information Technology Application | **NPTEL:** Generative AI For Computer Vision (2026) – `Elite + Gold`